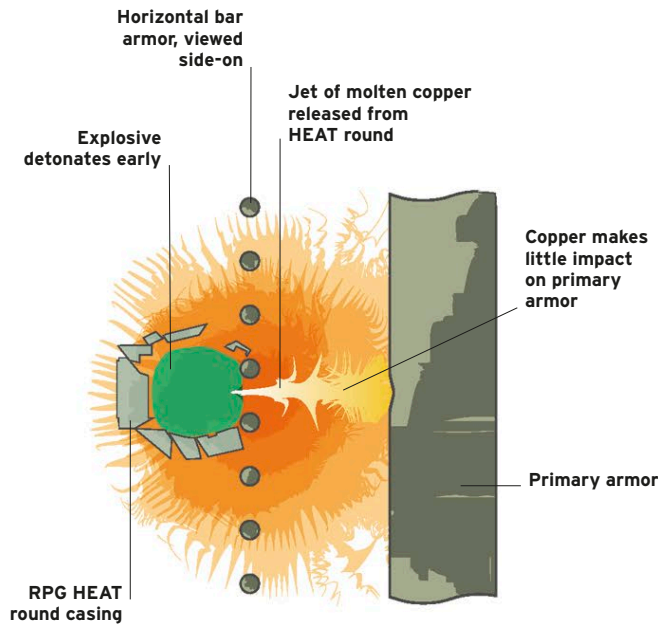


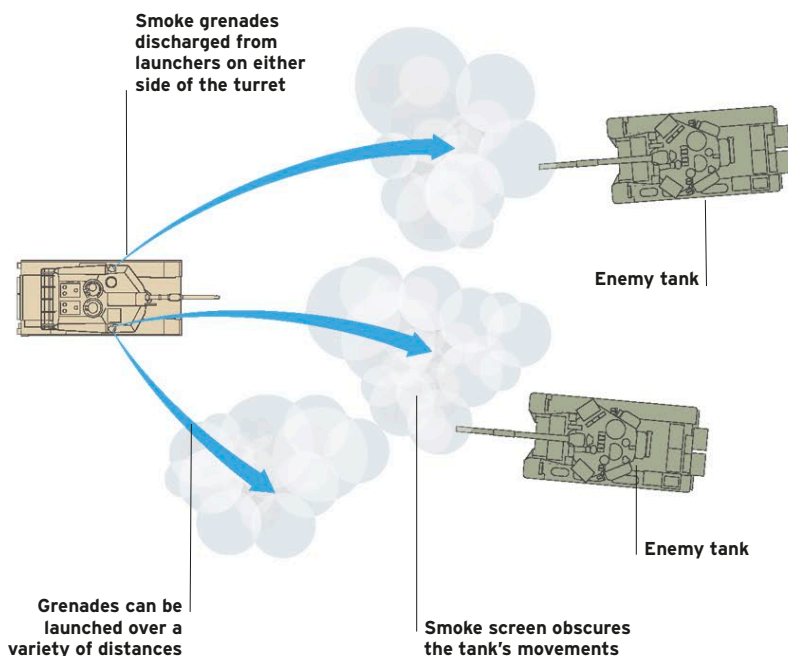
BAR ARMOR

Fitting light armored vehicles with bar armor is an inexpensive way of improving their overall level of protection by mounting a framework of hardened steel bars (usually horizontally) over vulnerable areas. Such protection is ineffective against kinetic energy rounds such as APFSDS, and of only limited efficacy against HESH rounds, but it can defeat lightweight HEAT rounds such as those delivered by grenade launchers like the RPG-7, which such vehicles will often encounter—by detonating them before they reach the bodywork of the vehicle itself.



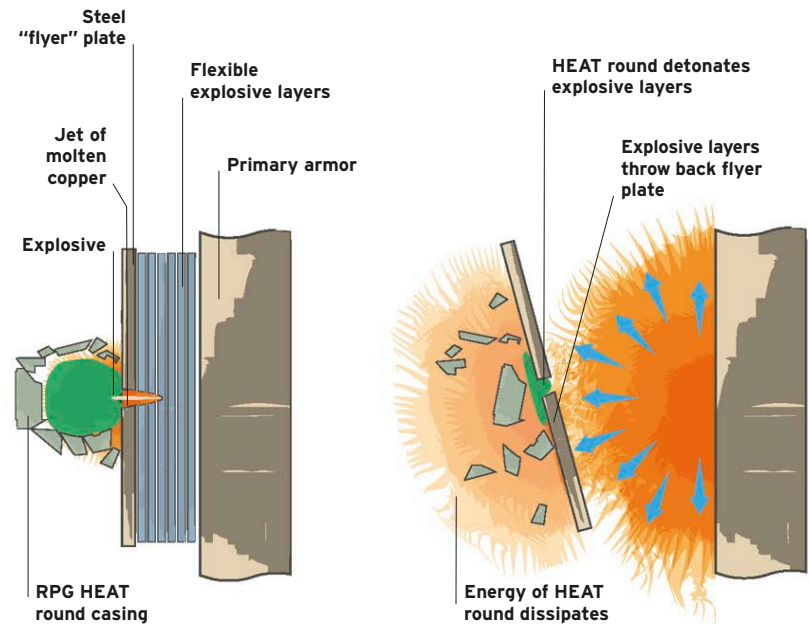
SMOKE GRENADES

Smoke has long been used to screen or obscure targets on the battlefield. Modern smoke grenades work in both the visual and infrared ends of the spectrum, meaning that tanks can also be hidden from thermal imaging systems. Since the 1940s, the method of choice for delivering smoke has been by means of grenades launched from projectors usually located on the vehicle's turret. These can be fired from within the vehicle, and a salvo of grenades will quickly form a large screen.



EXPLOSIVE REACTIVE ARMOR

An alternative form of supplementary armor, Explosive Reactive Armor (ERA) consists of a layer of relatively thin armor plating with a backing of high explosive. When the exposed armor plate is struck by a HEAT projectile, the jet of molten metal formed in the incoming round's warhead pierces it in the usual way, but then detonates the high explosive charge beneath, which reacts by blowing the entire panel off the target vehicle before the HEAT round can penetrate its main defensive armor. It can be defeated by so-called "tandem charge" HEAT rounds, which employ two charges, the second detonating milliseconds after the first, by which time the armor has been exposed.



CAMOUFLAGE

The intricate paint scheme on the first tanks was intended to hide them from enemy guns. Ever since, camouflage has become more sophisticated in order to defeat increasingly capable sensors. Methods have included paint, infrared suppressive paint, and thermal cladding.



Challenger 2 thermal insulation

The easiest way to detect a large vehicle in poor visibility is by means of its thermal image, or the heat it gives off. A surprising degree of protection can be achieved simply by equipping the vehicle with thermal insulation—such as the Solar Shield system fitted to this Challenger 2.



PL-01 radiation-absorbent coating

The experimental Polish PL-01 tank is coated with radiation-absorbent material, which "soaks up" all forms of electromagnetic radiation, including radar. This technology exists in a variety of types, and is similar to that used in so-called stealth aircraft.